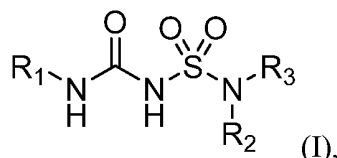


CLAIMS:

1. A compound of Formula (I):



or a ~~prodrug~~, solvate, or pharmaceutically acceptable salt thereof, wherein:

 R₁ is C₃-C₁₆ cycloalkyl, or C₅-C₁₀ aryl, wherein the C₃-C₈ monocyclic cycloalkyl, polycyclic cycloalkyl, or C₅-C₆ aryl is optionally substituted by one or more **R_{1s}**; wherein each **R_{1s}** is independently C₁-C₆ alkyl, C₁-C₆ haloalkyl, C₁-C₆ alkoxy, C₁-C₆ haloalkoxy, or halo;

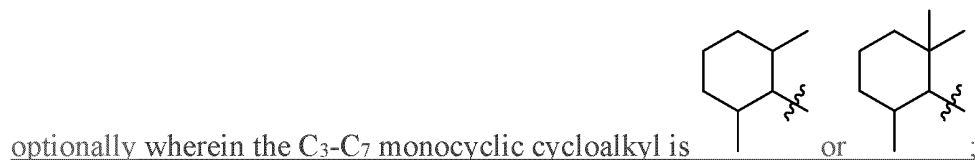
R₂ is -(C**X₂****X₂**)_n-**R_{2s}**, wherein n is 0, 1, or 2, and each **X₂** is independently H, C₁-C₆ alkyl, C₂-C₆ alkenyl, or C₂-C₆ alkynyl, wherein the C₁-C₆ alkyl, C₂-C₆ alkenyl, or C₂-C₆ alkynyl is optionally substituted with one or more halo, -CN, -OH, -O(C₁-C₆ alkyl), -NH₂, -NH(C₁-C₆ alkyl), -N(C₁-C₆ alkyl)₂, or oxo;

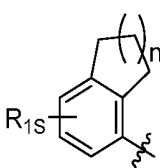
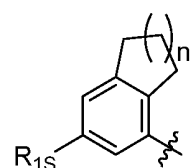
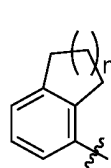
R_{2s} is 4- to 8-membered heterocycloalkyl optionally substituted with one or more C₁-C₆ alkyl, C₂-C₆ alkenyl, C₂-C₆ alkynyl, C₁-C₆ haloalkyl, halo, -CN, -OH, -O(C₁-C₆ alkyl), -NH₂, -NH(C₁-C₆ alkyl), -N(C₁-C₆ alkyl)₂, or oxo; and

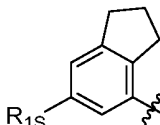
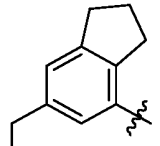
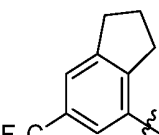
R₃ is 7- to 12-membered heterocycloalkyl or 5- or 6-membered heteroaryl optionally substituted with one or more **R_{3s}**, wherein each **R_{3s}** is independently C₁-C₆ alkyl, C₁-C₆ haloalkyl, C₃-C₈ cycloalkyl, halo, or C₃-C₈ heterocycloalkyl wherein the C₁-C₆ alkyl, C₁-C₆ haloalkyl, C₃-C₈ cycloalkyl, or C₃-C₈ heterocycloalkyl is optionally substituted with -O(C₁-C₆ alkyl), -N(C₁-C₆ alkyl)₂, halo, or -CN.

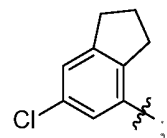
2. The compound of any one of the preceding claims, wherein **R₁** is C₃-C₇ monocyclic cycloalkyl, C₉-C₁₀ bicyclic cycloalkyl, C₁₂-C₁₆ tricyclic cycloalkyl, or C₅-C₁₀ aryl, wherein the C₃-C₇ monocyclic cycloalkyl, C₉-C₁₀ bicyclic cycloalkyl, C₁₂-C₁₆ tricyclic cycloalkyl, or C₅-C₁₀ aryl, is optionally substituted by one or more **R_{1s}**~~C₉-C₁₀ bicyclic cycloalkyl;~~
optionally wherein

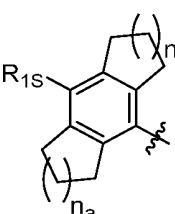
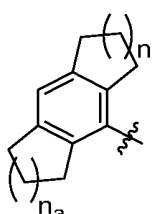
(a) the C₃-C₇ monocyclic cycloalkyl is cyclopentyl, cyclohexyl, or cycloheptyl, wherein the cyclopentyl, cyclohexyl, or cycloheptyl is optionally substituted by one or more R_{1S}; further

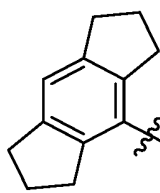


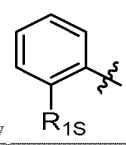
(b) the C₉-C₁₀ bicyclic cycloalkyl is , , or , wherein n

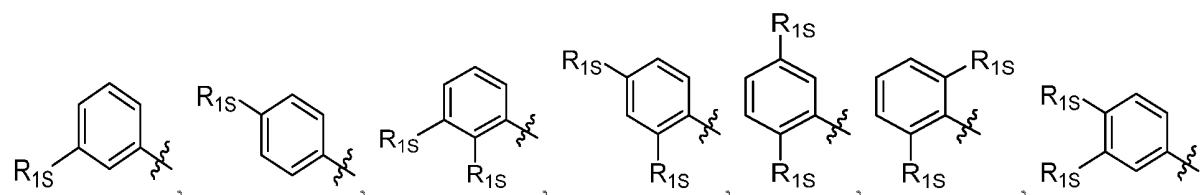
is 0, 1, or 2, optionally ; further optionally , F₃C-, or

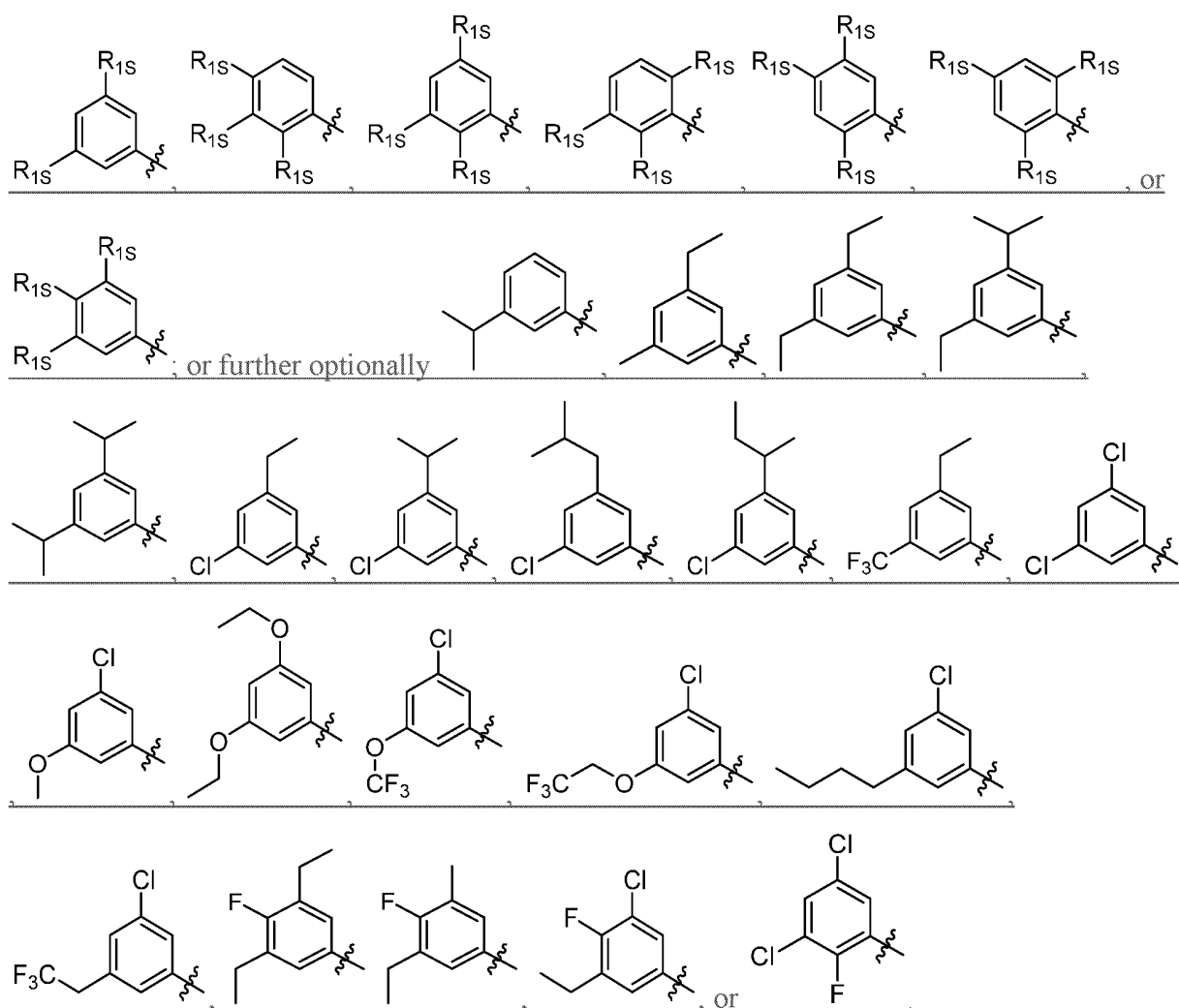


(c) the C₁₂-C₁₆ tricyclic cycloalkyl is , or , wherein n and n_a

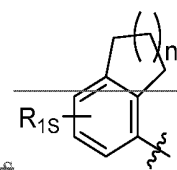
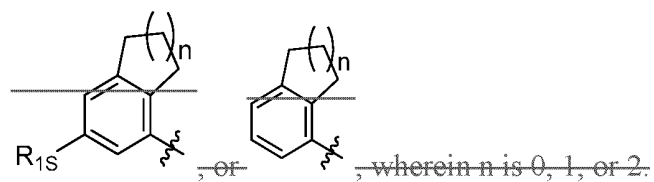
each independently are 0, 1, 2, or 3; further optionally ; and/or

(d) the C₅-C₁₀ aryl is phenyl is substituted by one, two, or three R_{1S}; optionally 

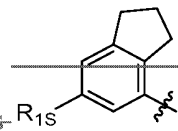




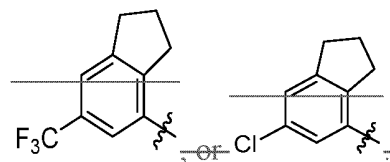
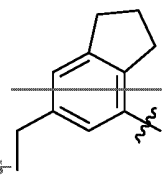
~~2. The compound of any one of the preceding claims, wherein R_4 is~~



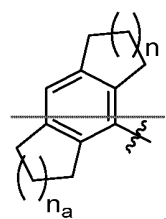
3. The compound of any one of the preceding claims, wherein R_1 is



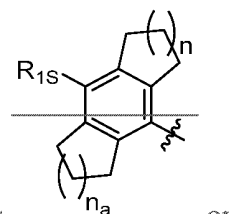
4. The compound of any one of the preceding claims, wherein R_1 is



5. The compound of any one of the preceding claims 1, wherein R_1 is C_{12} - C_{16} tricyclic cycloalkyl.

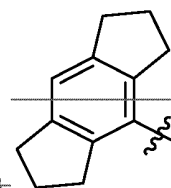


6. The compound of any one of the preceding claims, wherein R_1 is



, wherein n and n_a each independently are 0, 1, 2, or 3.

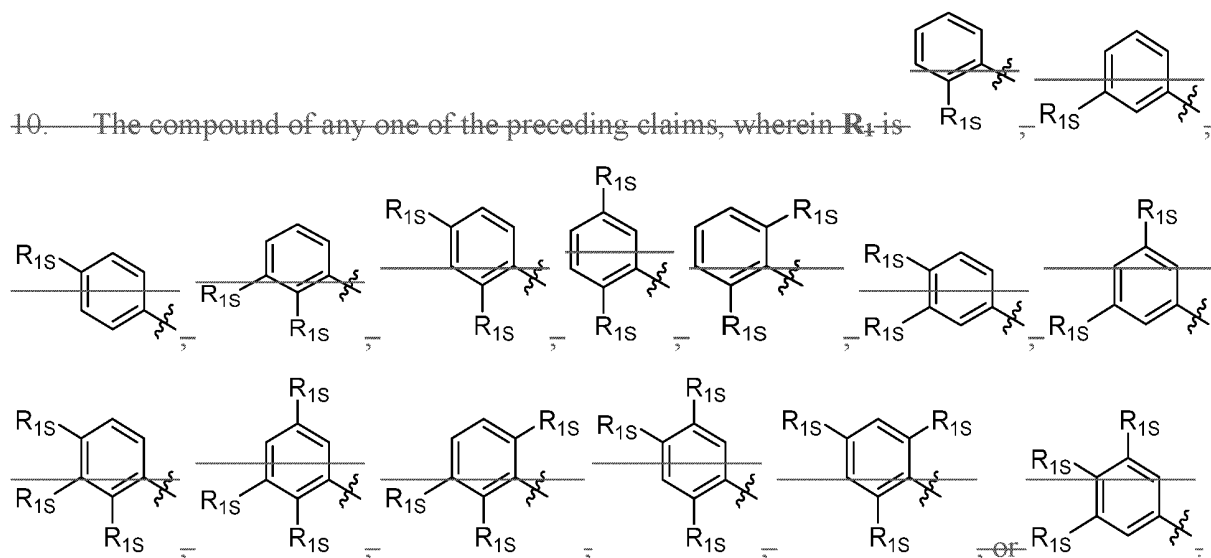
7. The compound of any one of the preceding claims 1, wherein R_1 is



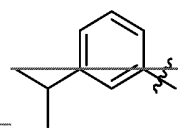
~~8. The compound of any one of the preceding claims, wherein, **R₄** is C₅-C₁₀-aryl optionally substituted by one or more **R₁₅**.~~

9. ~~The compound of any one of the preceding claims, wherein **R**₄ is phenyl is substituted by one, two, or three **R**₁₅.~~

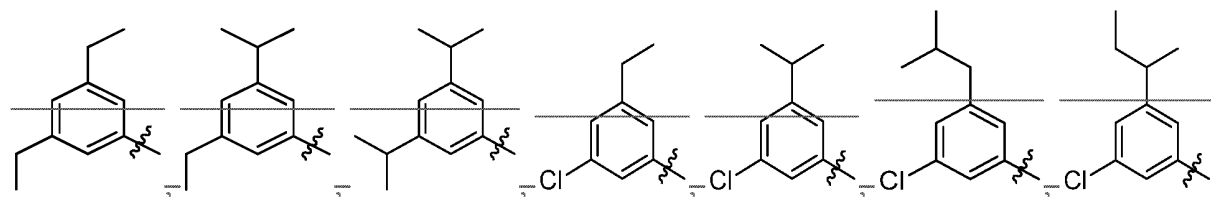
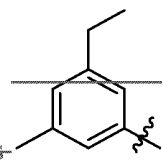
~~10. The compound of any one of the preceding claims, wherein R_4 is~~

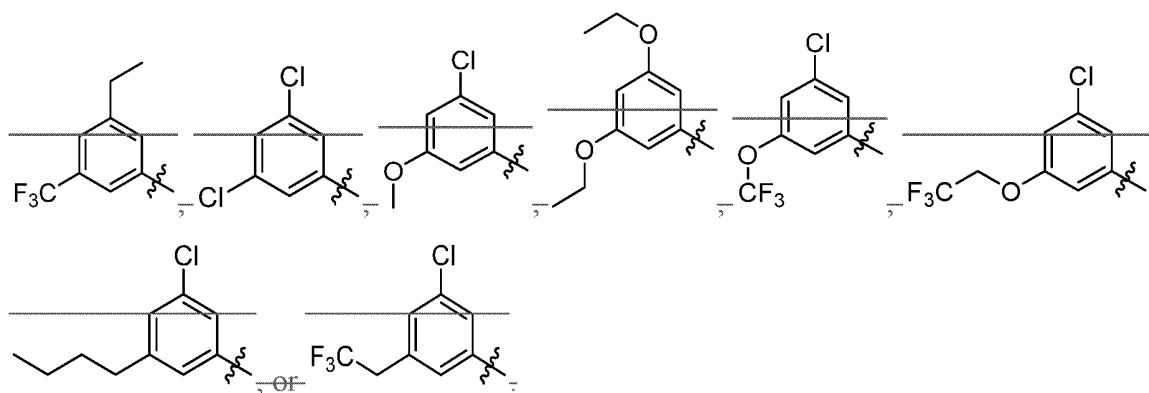


~~11. The compound of any one of the preceding claims, wherein R_4 is~~

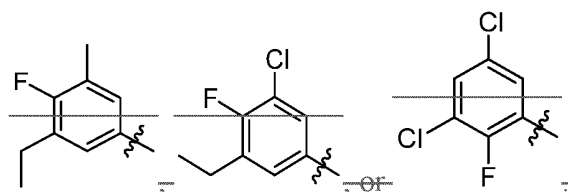
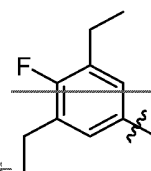


~~12. The compound of any one of the preceding claims, wherein R_4 is~~

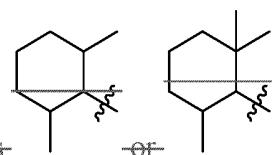




13. The compound of any one of the preceding claims, wherein R_4 is



14. The compound of any one of the preceding claims, wherein R_4 is cyclopentyl, cyclohexyl, or cycloheptyl, wherein the cyclopentyl, cyclohexyl, or cycloheptyl is optionally substituted by one or more R_{15} :



15. The compound of any one of the preceding claims, wherein R_4 is

16. The compound of any one of the preceding claims, wherein at least one R_{15} is methyl, ethyl, isopropyl, isobutyl, secbutyl, methoxy, ethoxy, CF_3 , OCF_3 , OCH_2CF_3 , F, or Cl.

17. The compound of any one of the preceding claims 1 to 3 or 2, wherein R_2 is $-R_{2S-3}$.

18.

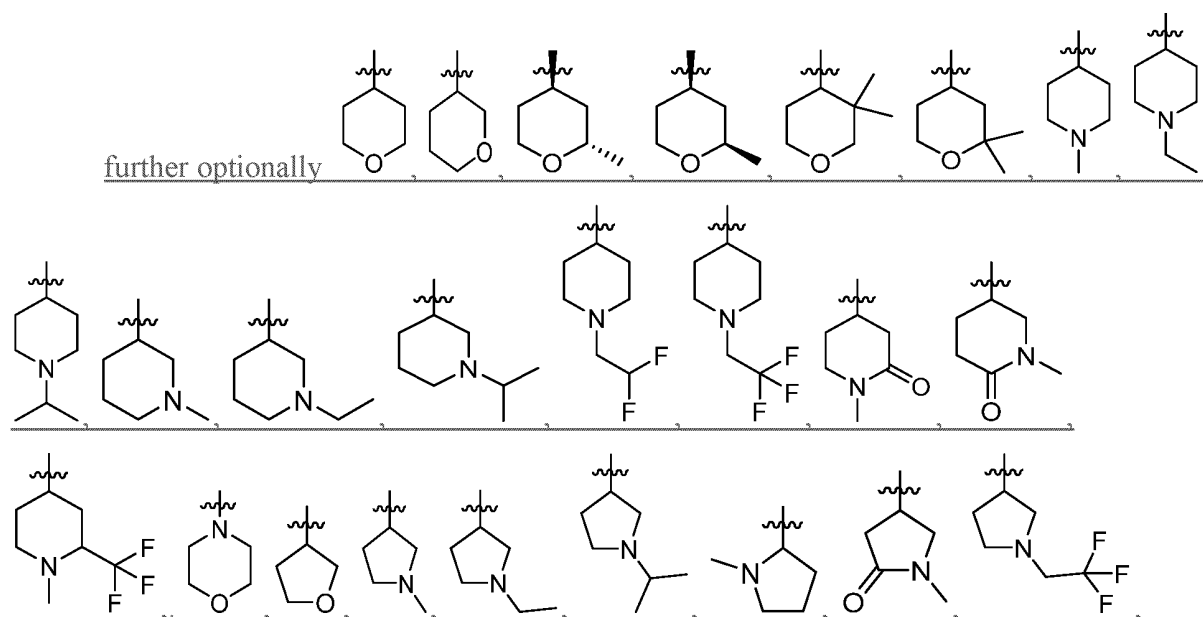
19. The compound of any one of the preceding claims, wherein R_2 is $-(CX_2X_2)-R_{2S-3}$ or

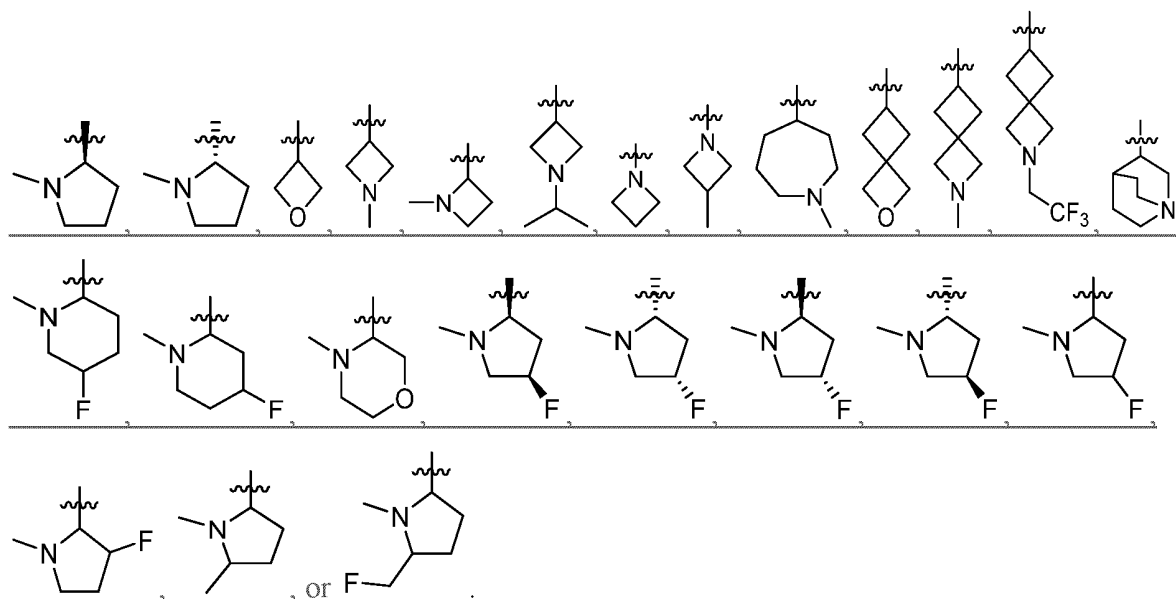
20. —

21.3. ~~The compound of any one of the preceding claims, wherein R_2 is $-(CX_2X_2)_2-R_{2S}$.~~22. ~~The compound of any one of the preceding claims, wherein each X_2 is H.~~23. ~~In some embodiments, at least one X_2 is C_1-C_6 alkyl, C_2-C_6 alkenyl, or C_2-C_6 alkynyl, wherein the C_1-C_6 alkyl, C_2-C_6 alkenyl, or C_2-C_6 alkynyl is optionally substituted with one or more halo, CN, OH, $O(C_1-C_6$ alkyl), NH_2 , $NH(C_1-C_6$ alkyl), $N(C_1-C_6$ alkyl) $_2$, or oxo.~~

4. ~~The compound of any one of the preceding claims 1 to 3, wherein R_{2S} is 4- to 8-membered heterocycloalkyl optionally substituted with one or more C_1-C_6 alkyl, C_2-C_6 alkenyl, C_2-C_6 alkynyl, C_1-C_6 haloalkyl, halo, -CN, -OH, $-O(C_1-C_6$ alkyl), $-NH_2$, $-NH(C_1-C_6$ alkyl), $-N(C_1-C_6$ alkyl) $_2$, or oxo;~~

~~optionally oxetanyl, azetidiny, tetrahydrofuranyl, pyrrolidinyl, tetrahydropyranyl, piperidinyl, morpholinyl, azepanyl, 2-azaspiro[3.3]heptane, 2-oxaspiro[3.3]heptane, or bridged quinuclidinyl, wherein the oxetanyl, azetidiny, tetrahydrofuranyl, pyrrolidinyl, tetrahydropyranyl, piperidinyl, morpholinyl, azepanyl, 2-azaspiro[3.3]heptane, 2-oxaspiro[3.3]heptane, or bridged quinuclidinyl is optionally substituted with one or more C_1-C_6 alkyl, C_1-C_6 haloalkyl, or oxo; or~~





24. The compound of any one of the preceding claims, wherein R_{2S} is 4- to 8-membered heterocycloalkyl.

25. The compound of any one of the preceding claims, wherein R_{2S} is 5- to 8-membered heterocycloalkyl.

26. The compound of any one of the preceding claims, wherein R_{2S} is 5- to 7-membered heterocycloalkyl.

27. The compound of any one of the preceding claims, wherein R_{2S} is 5- to 6-membered heterocycloalkyl.

28. The compound of any one of the preceding claims, wherein R_{2S} is 6-membered heterocycloalkyl optionally substituted with one or more C_1 - C_6 alkyl, C_2 - C_6 alkenyl, C_2 - C_6 alkynyl, C_1 - C_6 haloalkyl, halo, CN, OH, $O(C_1$ - C_6 alkyl), NH_2 , $NH(C_1$ - C_6 alkyl), $N(C_1$ - C_6 alkyl) $_2$, or oxo.

29. The compound of any one of the preceding claims, wherein R_{2S} is tetrahydropyranyl optionally substituted with one or more C_1 - C_6 alkyl, C_1 - C_6 haloalkyl, or oxo.

30. — ~~The compound of any one of the preceding claims, wherein R_{2S} is piperidinyl optionally substituted with one or more C_1-C_6 alkyl, C_1-C_6 haloalkyl, or oxo.~~

31. — ~~The compound of any one of the preceding claims, wherein R_{2S} is morpholinyl optionally substituted with one or more C_1-C_6 alkyl, C_1-C_6 haloalkyl, or oxo.~~

32. — ~~The compound of any one of the preceding claims, wherein R_{2S} is 5-membered heterocycloalkyl optionally substituted with one or more C_1-C_6 alkyl, C_2-C_6 alkenyl, C_2-C_6 alkynyl, C_1-C_6 haloalkyl, halo, CN, OH, $O(C_1-C_6$ alkyl), NH_2 , $NH(C_1-C_6$ alkyl), $N(C_1-C_6$ alkyl)₂, or oxo.~~

33. — ~~The compound of any one of the preceding claims, wherein R_{2S} is tetrahydrofuranyl optionally substituted with one or more C_1-C_6 alkyl, C_1-C_6 haloalkyl, or oxo.~~

34. — ~~The compound of any one of the preceding claims, wherein R_{2S} is pyrrolidinyl optionally substituted with one or more C_1-C_6 alkyl, C_1-C_6 haloalkyl, or oxo.~~

35. — ~~The compound of any one of the preceding claims, wherein R_{2S} is 4-membered heterocycloalkyl optionally substituted with one or more C_1-C_6 alkyl, C_2-C_6 alkenyl, C_2-C_6 alkynyl, C_1-C_6 haloalkyl, halo, CN, OH, $O(C_1-C_6$ alkyl), NH_2 , $NH(C_1-C_6$ alkyl), $N(C_1-C_6$ alkyl)₂, or oxo.~~

36. — ~~The compound of any one of the preceding claims, wherein R_{2S} is oxetanyl optionally substituted with one or more C_1-C_6 alkyl, C_1-C_6 haloalkyl, or oxo.~~

37. — ~~The compound of any one of the preceding claims, wherein R_{2S} is azetidiny optionally substituted with one or more C_1-C_6 alkyl, C_1-C_6 haloalkyl, or oxo.~~

38. — ~~The compound of any one of the preceding claims, wherein R_{2S} is 7-membered heterocycloalkyl optionally substituted with one or more C_1-C_6 alkyl, C_2-C_6 alkenyl, C_2-C_6~~

alkynyl, C₄-C₆ haloalkyl, halo, ~~CN, OH, O(C₄-C₆ alkyl), NH₂, NH(C₄-C₆ alkyl), N(C₄-C₆ alkyl)₂, or oxo.~~

39. ~~The compound of any one of the preceding claims, wherein R_{2S} is azepanyl optionally substituted with one or more C₄-C₆ alkyl, C₄-C₆ haloalkyl, or oxo.~~

40. ~~The compound of any one of the preceding claims, wherein R_{2S} is 8-membered heterocycloalkyl optionally substituted with one or more C₄-C₆ alkyl, C₂-C₆ alkenyl, C₂-C₆ alkynyl, C₄-C₆ haloalkyl, halo, ~~CN, OH, O(C₄-C₆ alkyl), NH₂, NH(C₄-C₆ alkyl), N(C₄-C₆ alkyl)₂, or oxo.~~~~

41. ~~The compound of any one of the preceding claims, wherein R_{2S} is spiro 7-membered heterocycloalkyl optionally substituted with one or more C₄-C₆ alkyl, C₄-C₆ haloalkyl, or oxo.~~

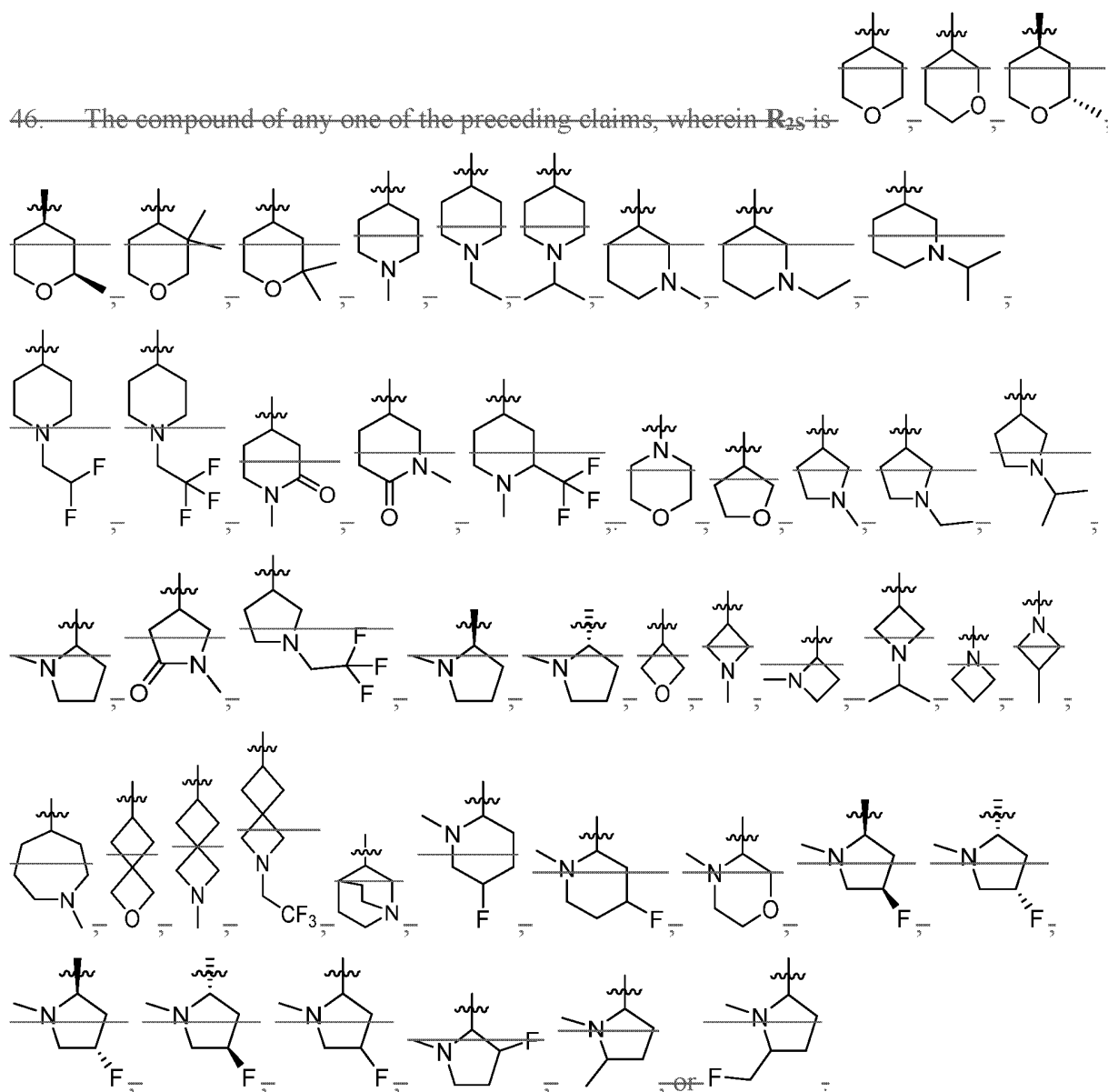
42. ~~The compound of any one of the preceding claims, wherein R_{2S} is 2-azaspiro[3.3]heptane optionally substituted with one or more C₄-C₆ alkyl, C₄-C₆ haloalkyl, or oxo.~~

43. ~~The compound of any one of the preceding claims, wherein R_{2S} is 2-oxaspiro[3.3]heptane optionally substituted with one or more C₄-C₆ alkyl, C₄-C₆ haloalkyl, or oxo.~~

44. ~~The compound of any one of the preceding claims, wherein R_{2S} is bridged heterocycloalkyl optionally substituted with one or more C₄-C₆ alkyl, C₂-C₆ alkenyl, C₂-C₆ alkynyl, C₄-C₆ haloalkyl, halo, ~~CN, OH, O(C₄-C₆ alkyl), NH₂, NH(C₄-C₆ alkyl), N(C₄-C₆ alkyl)₂, or oxo.~~~~

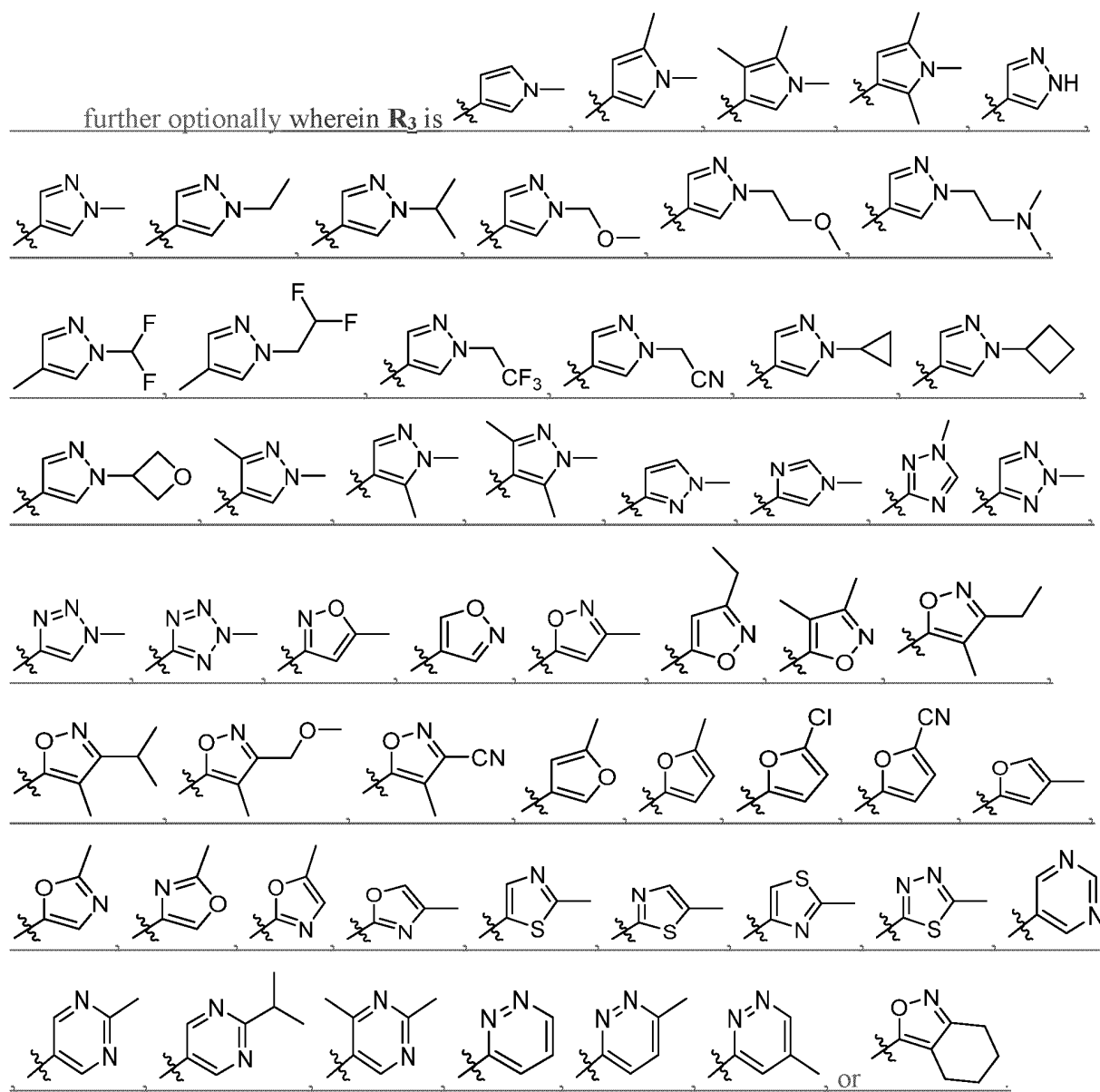
45. ~~The compound of any one of the preceding claims, wherein R_{2S} is bridged quinuclidinyl optionally substituted with one or more C₄-C₆ alkyl, C₄-C₆ haloalkyl, or oxo.~~

46. ~~The compound of any one of the preceding claims, wherein R_{25} is~~

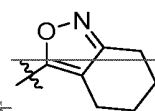


5. ~~The compound of any one of the preceding claims 1 to 4, wherein R_3 is 7- to 12-~~
 membered heterocycloalkyl optionally substituted with one or more R_{3S}

~~optionally wherein R_3 is pyrrolyl, pyrazolyl, imidazolyl, triazolyl, tetrazolyl, isoxazolyl, furanyl, oxazolyl, isothiazolyl, thiadiazolyl, pyridinyl, pyrimidinyl, pyridazinyl, or 4,5,6,7-tetrahydrobenzo[c]isoxazole, wherein the- pyrrolyl, pyrazolyl, imidazolyl, triazolyl, tetrazolyl, isoxazolyl, furanyl, oxazolyl, isothiazolyl, thiadiazolyl, pyridinyl, pyrimidinyl, pyridazinyl, or 4,5,6,7-tetrahydrobenzo[c]isoxazole is optionally substituted with one or more R_{3S} ; or~~



~~47. The compound of any one of the preceding claims, wherein R₃ is~~



~~48. The compound of any one of the preceding claims, wherein **R**₃ is 5- or 6-membered heteroaryl optionally substituted with one or more **R**_{3s}, wherein each **R**_{3s} is independently C₁-C₆ alkyl, C₁-C₆ haloalkyl, C₃-C₈ cycloalkyl, halo, or C₂-C₈ heterocycloalkyl wherein the C₁-C₆~~

alkyl, C₄-C₆ haloalkyl, C₃-C₈ cycloalkyl, or C₃-C₈ heterocycloalkyl is optionally substituted with ~~O(C₄-C₆ alkyl), N(C₄-C₆ alkyl)₂, halo, or CN.~~

49. ~~The compound of any one of the preceding claims, wherein R₃ is 5-membered heteroaryl optionally substituted with one or more R_{3S}, wherein each R_{3S} is independently C₄-C₆ alkyl, C₄-C₆ haloalkyl, C₃-C₈ cycloalkyl, halo, or C₃-C₈ heterocycloalkyl wherein the C₄-C₆ alkyl, C₄-C₆ haloalkyl, C₃-C₈ cycloalkyl, or C₃-C₈ heterocycloalkyl is optionally substituted with O(C₄-C₆ alkyl), N(C₄-C₆ alkyl)₂, halo, or CN.~~

50. ~~The compound of any one of the preceding claims, wherein R₃ is 6-membered heteroaryl optionally substituted with one or more R_{3S}, wherein each R_{3S} is independently C₄-C₆ alkyl, C₄-C₆ haloalkyl, C₃-C₈ cycloalkyl, halo, or C₃-C₈ heterocycloalkyl wherein the C₄-C₆ alkyl, C₄-C₆ haloalkyl, C₃-C₈ cycloalkyl, or C₃-C₈ heterocycloalkyl is optionally substituted with O(C₄-C₆ alkyl), N(C₄-C₆ alkyl)₂, halo, or CN.~~

51. ~~The compound of any one of the preceding claims, wherein R₃ is pyrrolyl optionally substituted by one or more R_{3S}.~~

52. ~~The compound of any one of the preceding claims, wherein R₃ is pyrazolyl optionally substituted by one or more R_{3S}.~~

53. ~~The compound of any one of the preceding claims, wherein R₃ is imidazolyl optionally substituted by one or more R_{3S}.~~

54. ~~The compound of any one of the preceding claims, wherein R₃ is triazolyl optionally substituted by one or more R_{3S}.~~

55. ~~The compound of any one of the preceding claims, wherein R₃ is tetrazolyl optionally substituted by one or more R_{3S}.~~

56.—The compound of any one of the preceding claims, wherein R_3 is isoxazolyl optionally substituted by one or more R_{3S} .

57.—The compound of any one of the preceding claims, wherein R_3 is furanyl optionally substituted by one or more R_{3S} .

58.—The compound of any one of the preceding claims, wherein R_3 is oxazolyl optionally substituted by one or more R_{3S} .

59.—The compound of any one of the preceding claims, wherein R_3 is isothiazolyl optionally substituted by one or more R_{3S} .

60.—The compound of any one of the preceding claims, wherein R_3 is thiadiazolyl optionally substituted by one or more R_{3S} .

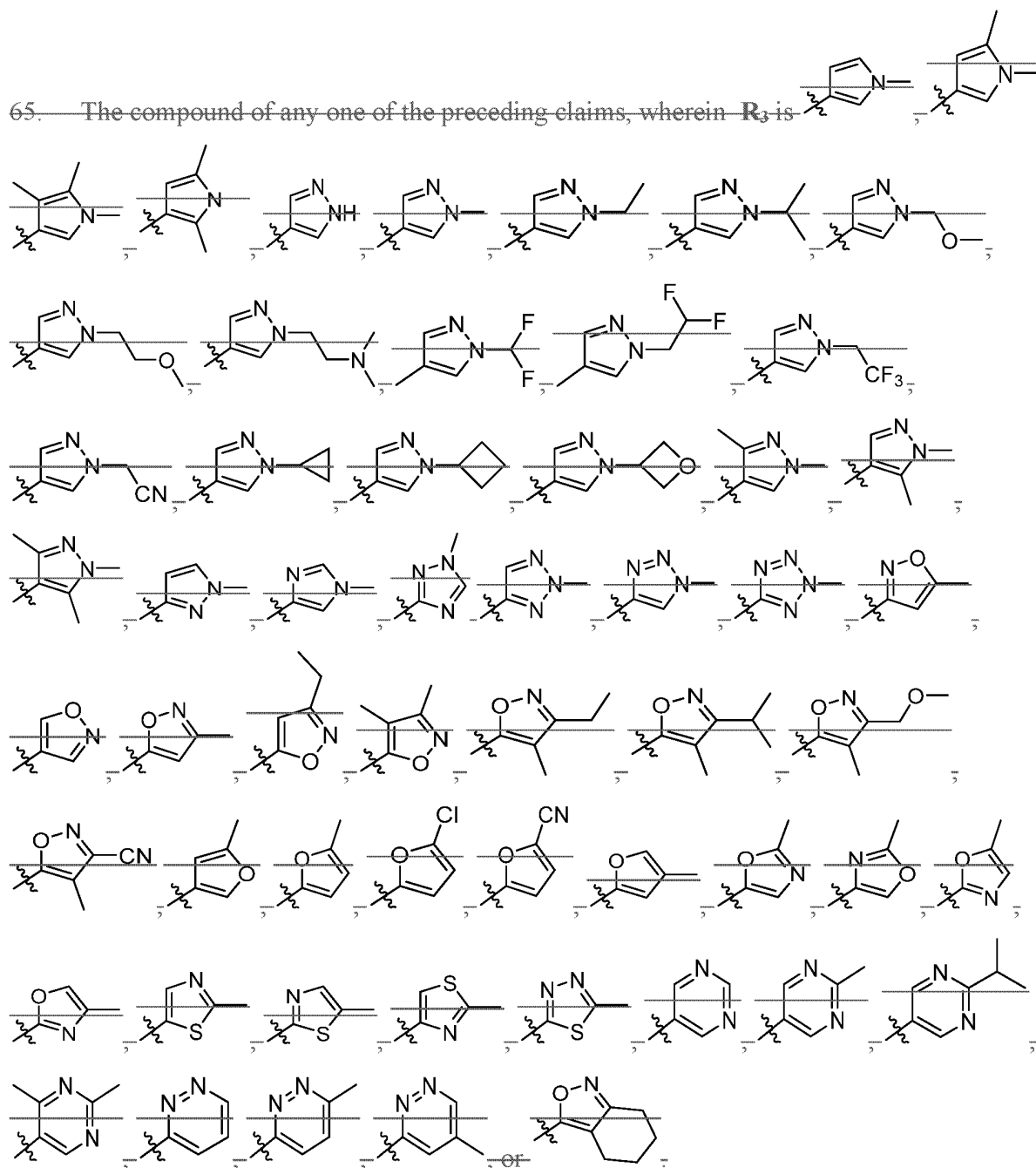
61.—The compound of any one of the preceding claims, wherein R_3 is pyridinyl optionally substituted by one or more R_{3S} .

62.—The compound of any one of the preceding claims, wherein R_3 is pyrimidinyl optionally substituted by one or more R_{3S} .

63.—The compound of any one of the preceding claims, wherein R_3 is pyridazinyl optionally substituted by one or more R_{3S} .

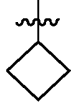
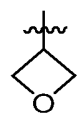
64.—The compound of any one of the preceding claims, wherein R_3 is 4,5,6,7-tetrahydrobenzo[c]isoxazole optionally substituted by one or more R_{3S} .

~~65. The compound of any one of the preceding claims, wherein R_3 is~~

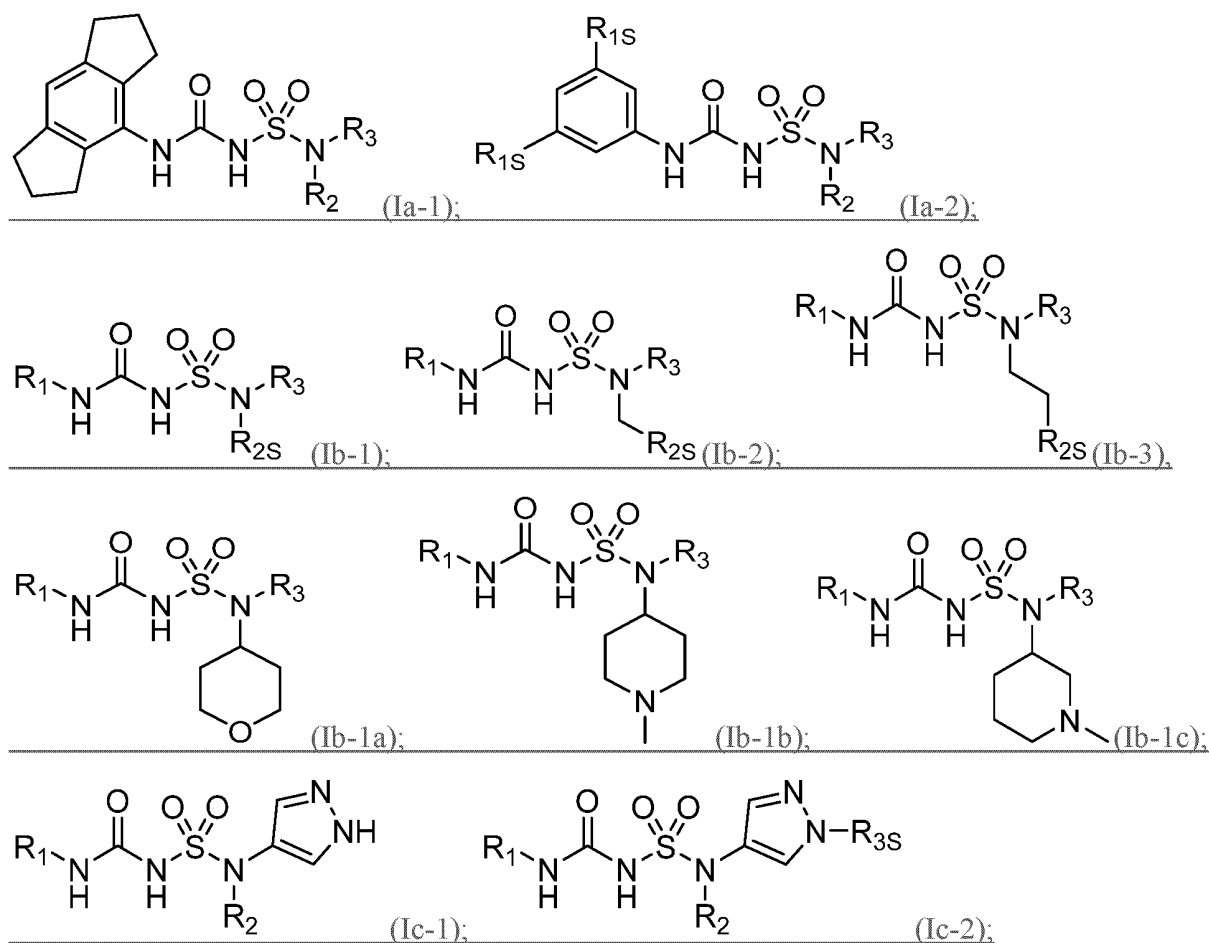


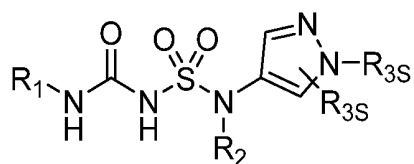
6. The compound of any one of the preceding claims 1 to 5, wherein at least one **R₃₅** is C₁-C₆ alkyl, C₁-C₆ haloalkyl, C₃-C₈ cycloalkyl, halo, or C₃-C₈ heterocycloalkyl wherein the C₁-C₆ alkyl, C₁-C₆ haloalkyl, C₃-C₈ cycloalkyl, or C₃-C₈ heterocycloalkyl is optionally substituted with -O(C₁-C₆ alkyl), -N(C₁-C₆ alkyl)₂, halo, or -CN;

optionally -

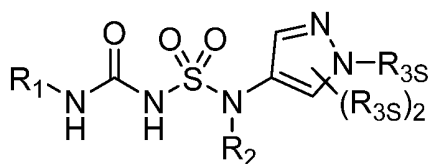
The compound of any one of the preceding claims, wherein at least one R_{3S} is , , or , methyl, ethyl, isopropyl, $-\text{CH}_2\text{OCH}_3$, $-\text{CH}_2\text{CF}_3$, $-\text{CH}_2\text{CH}_2\text{OCH}_3$, $-\text{CH}_2\text{CN}$, CH_2CF_2 , or $-\text{CH}_2\text{CH}_2\text{N}(\text{CH}_3)_2$.

7. The compound of any one of the preceding claims, wherein the compound is of Formula (Ia-1), (Ia-2), (Ib-1), (Ib-2), (Ib-3), (Ib-1a), (Ib-1b), (Ib-1c), (Ic-1), (Ic-2), (Ic-3), (Ic-4), (Ic-5), (Ic-6), (Ic-7), (Ic-8), (Iab-1), (Iab-2), (Iac-1), (Iac-2), (Iac-3), (Iac-4), (Ibc-1), (Ibc-2), (Iabc-1), (Iabc-1a), or (Iabc-1b):

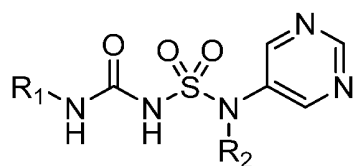




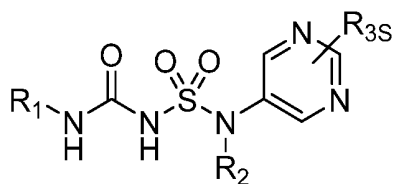
(Ic-3);



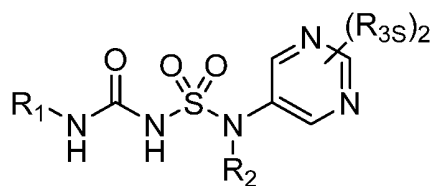
(Ic-4);



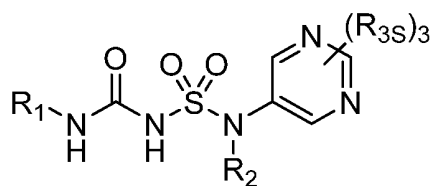
(Ic-5);



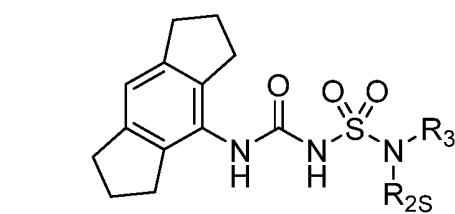
(Ic-6);



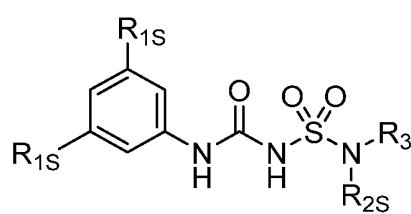
(Ic-7);



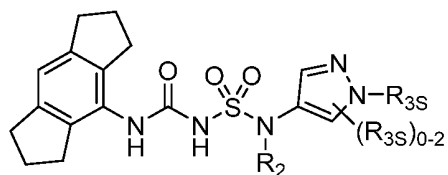
(Ic-8);



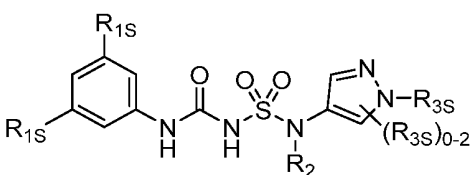
(Iab-1);



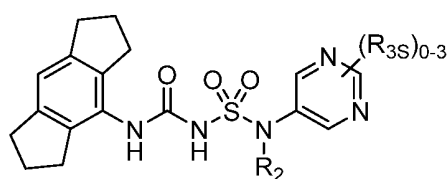
(Iab-2);



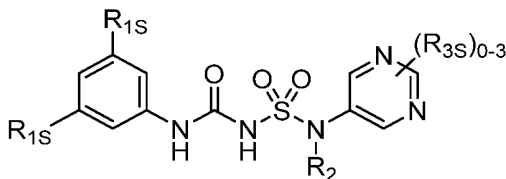
(Iac-1);



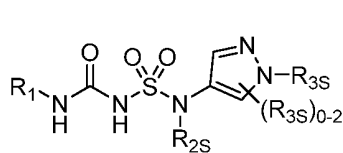
(Iac-2);



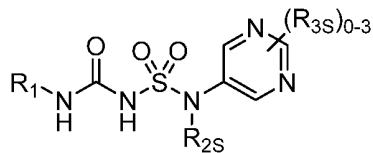
(Iac-3);



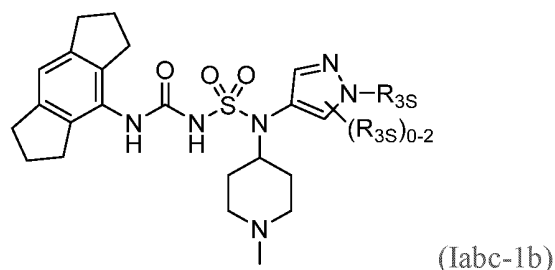
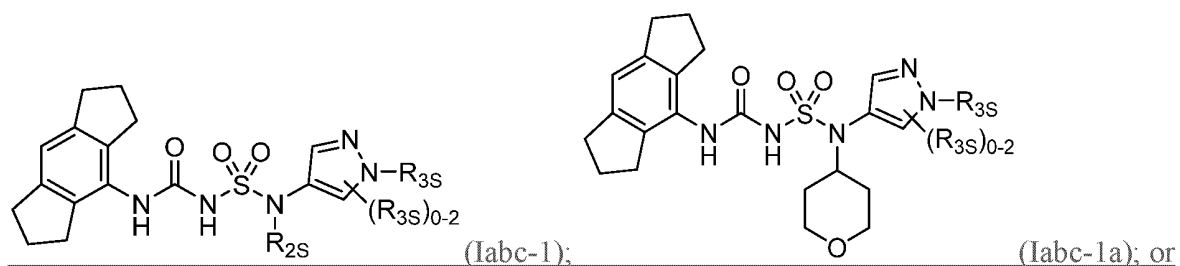
(Iac-4);



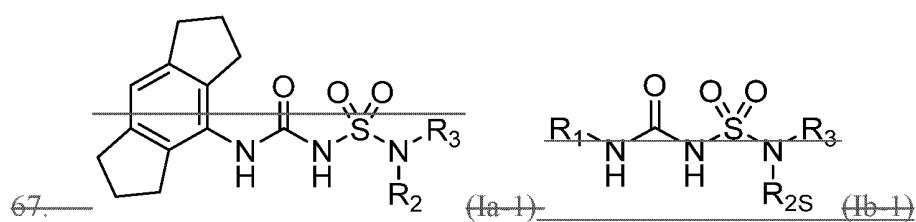
(Ibc-1);



(Ibc-2);

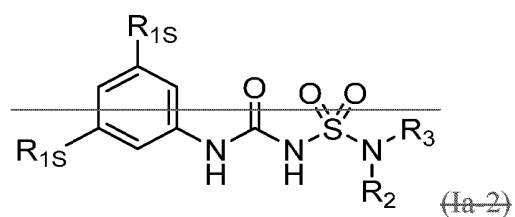


66. ~~(Ia-1) or (Ib-1):~~



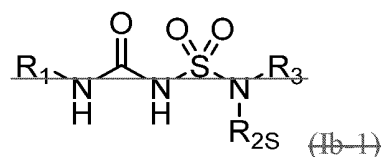
or a ~~prodrug~~, solvate, or pharmaceutically acceptable salt thereof, wherein **R₁**, **R_{2S}**, **R₂**, and **R₃** are as described herein.

68. ~~The compound of any one of the preceding claims, wherein the compound is of Formula (Ia-2):~~



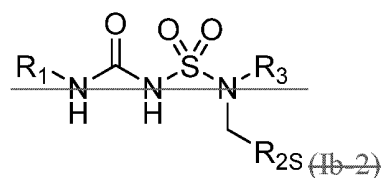
~~or a prodrug, solvate, or pharmaceutically acceptable salt thereof, wherein **R_{1S}**, **R₂** and **R₃** are as described herein.~~

69. ~~The compound of any one of the preceding claims, wherein the compound is of Formula (Ib-1):~~



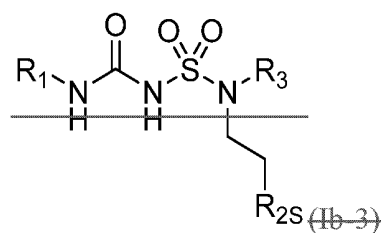
— or a prodrug, solvate, or pharmaceutically acceptable salt thereof, wherein **R₁**, **R_{2S}**, and **R₃** are as described herein.

70. — The compound of any one of the preceding claims, wherein the compound is of Formula (Ib-2):



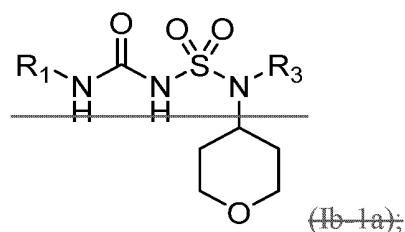
— or a prodrug, solvate, or pharmaceutically acceptable salt thereof, wherein **R₁**, **R_{2S}**, and **R₃** are as described herein.

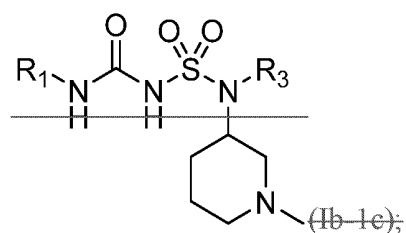
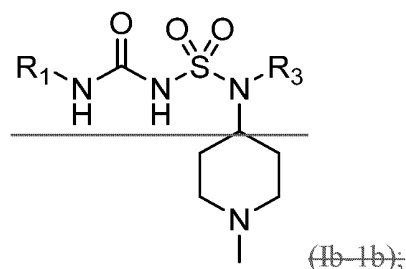
71. — The compound of any one of the preceding claims, wherein the compound is of Formula (Ib-3):



— or a prodrug, solvate, or pharmaceutically acceptable salt thereof, wherein **R₁**, **R_{2S}**, and **R₃** are as described herein.

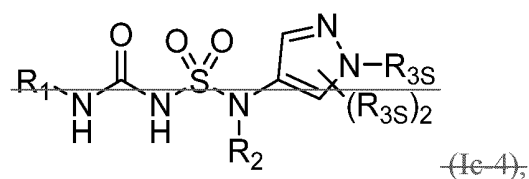
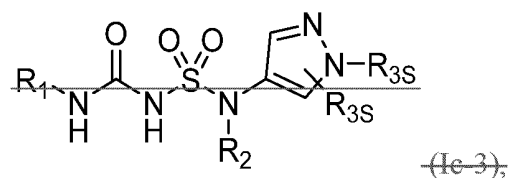
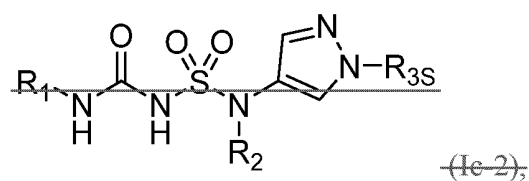
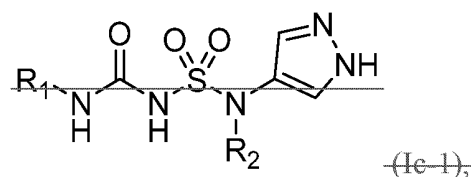
72. — The compound of any one of the preceding claims, wherein the compound is of any one of Formulae (Ib-1a), (Ib-1b), or (Ib-1c):

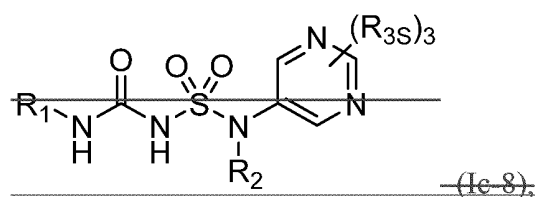
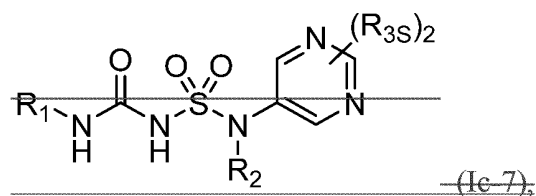
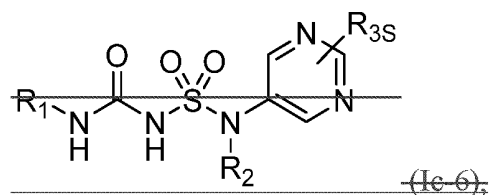
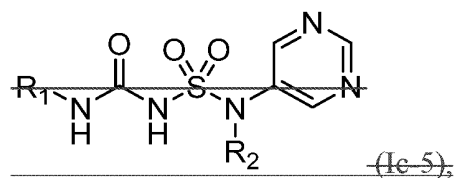




— or a prodrug, solvate, or pharmaceutically acceptable salt thereof, wherein **R₁** and **R₃** are as described herein.

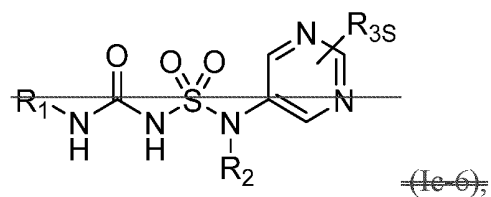
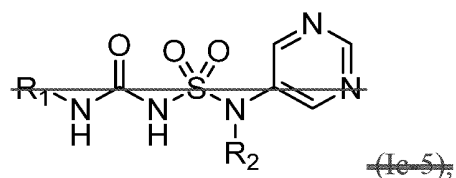
73. — The compound of any one of the preceding claims, wherein the compound is of Formula (Ic-1), (Ic-2), (Ic-3), or (Ic-4), (Ic-5), (Ic-6), (Ic-7) or (Ic-8):

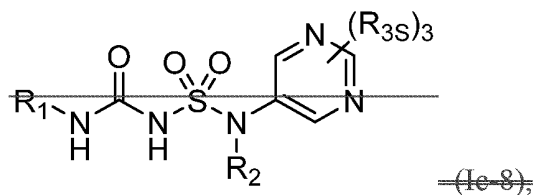
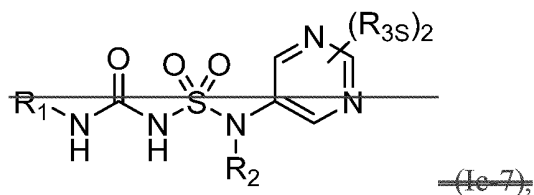




— or a prodrug, solvate, or pharmaceutically acceptable salt thereof, wherein **R₁**, **R₂** and **R_{3S}** are as described herein.

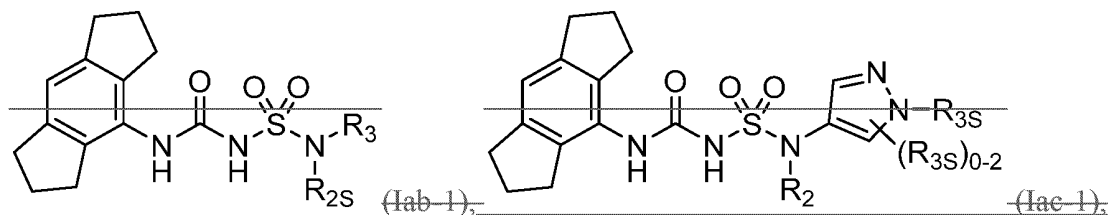
74. The compound of any one of the preceding claims, wherein the compound is of Formula (Ic-5), (Ic-6), (Ic-7), or (Ic-8):





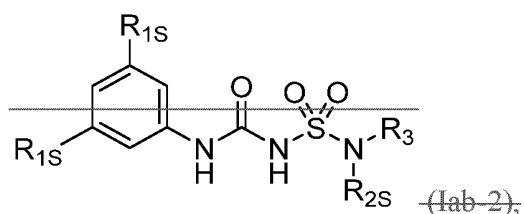
— or a prodrug, solvate, or pharmaceutically acceptable salt thereof, wherein ~~**R₁**~~, ~~**R₂**~~ and ~~**R_{3S}**~~, are as described herein.

75. — The compound of any one of the preceding claims, wherein the compound is of Formula ~~(Iab-1)~~ or ~~(Iac-1)~~:



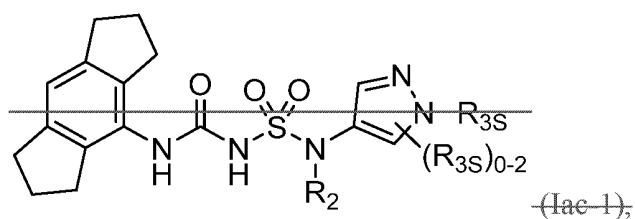
— or a prodrug, solvate, or pharmaceutically acceptable salt thereof, wherein ~~**R_{1S}**~~, ~~**R_{2S}**~~, ~~**R_{3S}**~~ and ~~**R₃**~~ are as described herein.

76. — The compound of any one of the preceding claims, wherein the compound is of Formula ~~(Iab-2)~~:



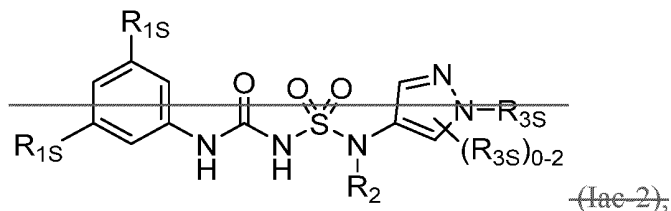
— or a prodrug, solvate, or pharmaceutically acceptable salt thereof, wherein ~~**R_{1S}**~~, ~~**R_{2S}**~~, and ~~**R₃**~~ are as described herein.

77. — The compound of any one of the preceding claims, wherein the compound is of Formula ~~(Iac-1)~~:



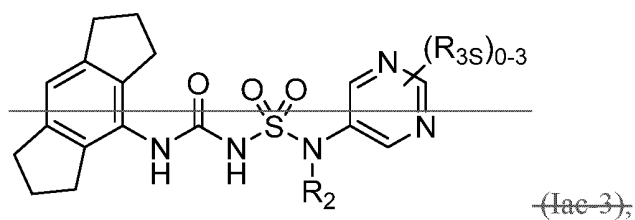
— or a prodrug, solvate, or pharmaceutically acceptable salt thereof, wherein **R₂** and **R_{3S}** are as described herein.

78. — The compound of any one of the preceding claims, wherein the compound is of Formula (Iac-2):



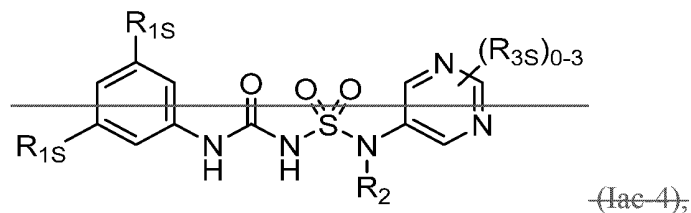
— or a prodrug, solvate, or pharmaceutically acceptable salt thereof, wherein **R_{1S}**, **R₂**, and **R_{3S}** are as described herein.

79. — The compound of any one of the preceding claims, wherein the compound is of Formula (Iac-3):



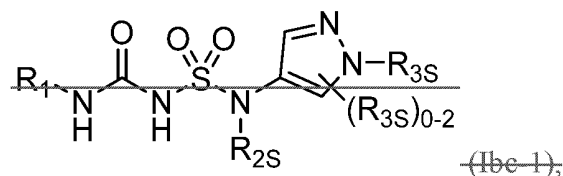
— or a prodrug, solvate, or pharmaceutically acceptable salt thereof, wherein **R₂** and **R_{3S}** are as described herein.

80. — The compound of any one of the preceding claims, wherein the compound is of Formula (Iac-4):



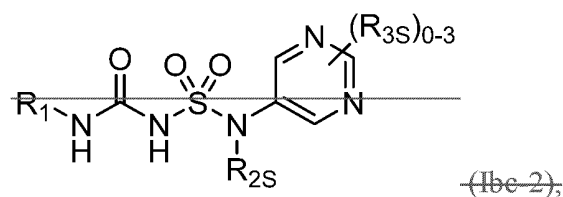
— or a prodrug, solvate, or pharmaceutically acceptable salt thereof, wherein **R_{1S}**, **R₂**, and **R_{3S}** are as described herein.

81. — The compound of any one of the preceding claims, wherein the compound is of Formula (Ibc-1):



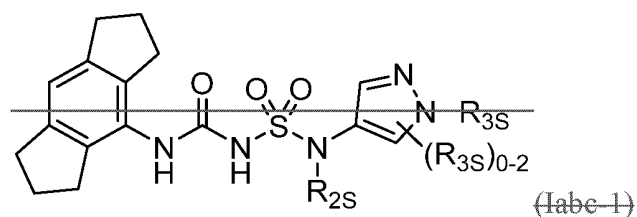
— or a prodrug, solvate, or pharmaceutically acceptable salt thereof, wherein **R₁**, **R_{2S}**, and **R_{3S}** are as described herein.

82. — The compound of any one of the preceding claims, wherein the compound is of Formula (Ibc-2):



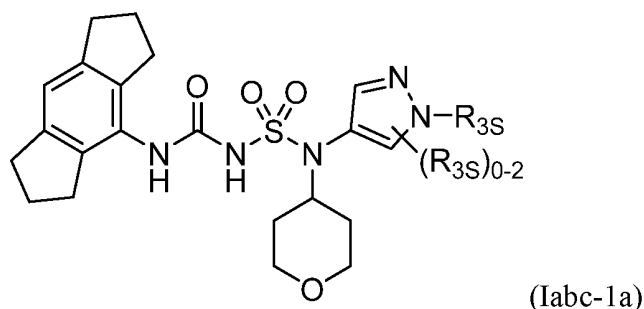
— or a prodrug, solvate, or pharmaceutically acceptable salt thereof, wherein **R₁**, **R_{2S}**, and **R_{3S}** are as described herein.

83. — The compound of any one of the preceding claims, wherein the compound is of Formula (Iabc-1):



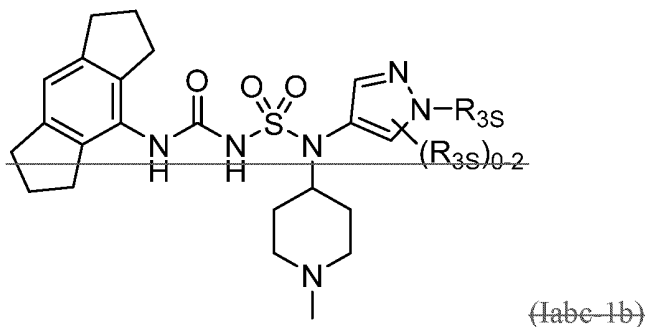
~~or a prodrug, solvate, or pharmaceutically acceptable salt thereof, wherein R_{2S} and R_{3S} are as described herein.~~

~~84.8.~~ The compound of any one of the preceding claims 1 to 7, wherein the compound is of Formula (Iabc-1a):



~~or a prodrug, solvate, or pharmaceutically acceptable salt thereof, wherein R_{3S} is as described herein.~~

~~85.~~ The compound of any one of the preceding claims, wherein the compound is of Formula (Iabc-1b):



~~or a prodrug, solvate, or pharmaceutically acceptable salt thereof, wherein R_{3S} is as described herein.~~

~~86.9.~~ The compound of any one of the claims 1- to 87~~158~~, being selected from Compound Nos. 1-166 and ~~prodrugs and~~ pharmaceutically acceptable salts thereof.

~~87.~~ The compound of any one of claims 1- 87, being selected from Compound Nos. 1- 166 and pharmaceutically acceptable salts thereof.

- ~~88. The compound of any one of claims 1-87, being selected from Compound Nos. 1-166.~~
- ~~89. The compound of any one of the claims 1-87, being selected from Compound Nos. 1A-166A and prodrugs and pharmaceutically acceptable salts thereof.~~
- ~~90. The compound of any one of claims 1-87, being selected from Compound Nos. 1A-166A and pharmaceutically acceptable salts thereof.~~
- ~~91. The compound of any one of claims 1-87, being selected from Compound Nos. 1A-166A.~~
- ~~92. A compound being an isotopic derivative of the compound of any one of claims 1-93.~~
- ~~93. The compound of claim 94, being a deuterium labeled compound.~~
- ~~94. The compound of claim 94, being a deuterium labeled compound of any one of Compound Nos. 1-166 and prodrugs and pharmaceutically acceptable salts thereof.~~
- ~~95. The compound of claim 94, being a deuterium labeled compound of any one of Compound Nos. 1-166.~~
- ~~96. The compound of claim 94, being a deuterium labeled compound of any one of Compound Nos. 1A-166A and prodrugs and pharmaceutically acceptable salts thereof.~~
- ~~97. The compound of claim 94, being a deuterium labeled compound of any one of Compound Nos. 1A-166A.~~
- ~~98. A compound obtainable by, or obtained by, a method described herein;
— optionally, the method comprises one or more steps described in Schemes 1-2.~~
- ~~99. A compound, by an intermediate obtained by a method for preparing the compound of any one of claims 1-93;~~

~~_____ optionally, the intermediate is selected from the intermediates described in Examples 1-166.~~

~~100.10. _____ A pharmaceutical composition comprising the compound of any one of claims 1- to 101 159 or a pharmaceutically acceptable salt thereof, and a pharmaceutically acceptable diluent or carrier.~~

~~101. The pharmaceutical composition of claim 102, wherein the compound is selected from Compound Nos. 1-166.~~

~~102. The pharmaceutical composition of claim 102, wherein the compound is selected from Compound Nos. 1A-166A.~~

~~103. _____~~

~~104. A method of inhibiting inflammasome activity, comprising contacting a cell with an effective amount of the compound of any one of claims 1-101 or a pharmaceutically acceptable salt thereof; optionally, the inflammasome is NLRP3 inflammasome, and the activity is *in vitro* or *in vivo*.~~

~~105. _____~~

~~106. A method of treating or preventing a disease or disorder in a subject in need thereof, comprising administering to the subject a therapeutically effective amount of the compound of any one of claims 1-101 or a pharmaceutically acceptable salt thereof, or the pharmaceutical composition of any one of claims 102-104.~~

~~107. _____~~

~~108. The method of claim 106, wherein the disease or disorder is associated with an implicated inflammasome activity; optionally, the disease or disorder is a disease or disorder in which inflammasome activity is implicated.~~

~~109. _____~~

~~110. The method of claim 106 or claim 107, wherein the disease or disorder is an inflammatory disorder, an autoinflammatory disorder, an autoimmune disorder, a neurodegenerative disease, or cancer.~~

~~111. _____~~

~~112. The method of any one of claims 106-108, wherein the disease or disorder is an inflammatory disorder, an autoinflammatory disorder or an autoimmune disorder; optionally, the disease or disorder is selected from cryopyrin-associated auto-inflammatory syndrome (CAPS; e.g., familial cold autoinflammatory syndrome (FCAS), Muckle-Wells syndrome (MWS), chronic infantile neurological cutaneous and articular (CINCA) syndrome/ neonatal-onset multisystem inflammatory disease (NOMID)), familial Mediterranean fever (FMF), nonalcoholic fatty liver disease (NAFLD), non-alcoholic steatohepatitis (NASH), gout, rheumatoid arthritis, osteoarthritis, Crohn's disease, chronic obstructive pulmonary disease (COPD), chronic kidney disease (CKD), fibrosis, obesity, type 2 diabetes, multiple sclerosis, dermatological disease (e.g., acne) and neuroinflammation occurring in protein misfolding diseases (e.g., Prion diseases).~~

~~113.—~~

~~114. The method of any one of claims 106-108, wherein disease or disorder is a neurodegenerative disease; optionally, the disease or disorder is Parkinson's disease or Alzheimer's disease.~~

~~115.—~~

~~116. The method of any one of claims 106-108, wherein the disease or disorder is cancer; optionally, the cancer is metastasising cancer, brain cancer, gastrointestinal cancer, skin cancer, non-small-cell lung carcinoma, head and neck squamous-cell carcinoma or colorectal adenocarcinoma.~~

~~117.11._____~~ The compound of any one of claims 1-~~to 99159~~, or the pharmaceutical composition of ~~any one of claims 102-104 1610~~, for use in inhibiting inflammasome activity; optionally, the inflammasome is NLRP3 inflammasome, and the activity is *in vitro* or *in vivo*.

~~118.—~~ The compound of any one of claims 1-~~to 99159~~, or the pharmaceutical composition of ~~any one of claims 102-104 1610~~, for use in treating or preventing a disease or disorder-; preferably

~~119.—~~

~~120.—~~ The compound or pharmaceutical composition of claim 113, wherein the disease or disorder is associated with an implicated inflammasome activity; optionally, the disease or disorder is a disease or disorder in which inflammasome activity is implicated.

~~121.—~~

~~12. The compound or pharmaceutical composition of claim 113 or claim 114,~~ wherein the disease or disorder is an inflammatory disorder, an autoinflammatory disorder, an autoimmune disorder, a neurodegenerative disease, or cancer.

13. The compound or pharmaceutical composition for use of any one of claims 11 or 123-115, wherein the disease or disorder is an inflammatory disorder, an autoinflammatory disorder or an autoimmune disorder; optionally, the disease or disorder is selected from cryopyrin-associated autoinflammatory syndrome (CAPS; e.g., familial cold autoinflammatory syndrome (FCAS), Muckle-Wells syndrome (MWS), chronic infantile neurological cutaneous and articular (CINCA) syndrome/ neonatal-onset multisystem inflammatory disease (NOMID)), familial Mediterranean fever (FMF), nonalcoholic fatty liver disease (NAFLD), non-alcoholic steatohepatitis (NASH), gout, rheumatoid arthritis, osteoarthritis, Crohn's disease, chronic obstructive pulmonary disease (COPD), chronic kidney disease (CKD), fibrosis, obesity, type 2 diabetes, multiple sclerosis, dermatological diseases (e.g., acne) and neuroinflammation occurring in protein misfolding diseases (e.g., Prion diseases).

14. The compound or pharmaceutical composition for use of any one of claims 113-11511 or 12, wherein disease or disorder is a neurodegenerative disease; optionally, the disease or disorder is Parkinson's disease or Alzheimer's disease.

15. The compound or pharmaceutical composition for use of any one of claims 113-11511 or 12, wherein the disease or disorder is cancer; optionally, the cancer is metastasising cancer, gastrointestinal cancer, brain cancer, skin cancer, non-small-cell lung carcinoma, head and neck squamous cell carcinoma or colorectal adenocarcinoma.

~~Use of the compound of any one of claims 1-99 or a pharmaceutically acceptable salt thereof in the manufacture of a medicament for inhibiting inflammasome activity; optionally, the inflammasome is NLRP3 inflammasome, and the activity is *in vitro* or *in vivo*.~~

~~Use of the compound of any one of claims 1-99 or a pharmaceutically acceptable salt thereof in~~

~~the manufacture of a medicament for treating or preventing a disease or disorder.~~

~~The use of claim 120, wherein the disease or disorder is associated with an implicated inflammasome activity; optionally, the disease or disorder is a disease or disorder in which inflammasome activity is implicated.~~

~~The use of claim 120 or claim 121, wherein the disease or disorder is an inflammatory disorder, an autoinflammatory disorder, an autoimmune disorder, a neurodegenerative disease, or cancer.~~

~~The use of any one of claims 120-122, wherein the disease or disorder is an inflammatory disorder, an autoinflammatory disorder or an autoimmune disorder; optionally, the disease or disorder is selected from cryopyrin-associated autoinflammatory syndrome (CAPS; e.g., familial cold autoinflammatory syndrome (FCAS), Muckle-Wells syndrome (MWS), chronic infantile neurological cutaneous and articular (CINCA) syndrome/ neonatal-onset multisystem inflammatory disease (NOMID)), familial Mediterranean fever (FMF), nonalcoholic fatty liver disease (NAFLD), non-alcoholic steatohepatitis (NASH), gout, rheumatoid arthritis, osteoarthritis, Crohn's disease, chronic obstructive pulmonary disease (COPD), chronic kidney disease (CKD), fibrosis, obesity, type 2 diabetes, multiple sclerosis, dermatological diseases (e.g., acne) and neuroinflammation occurring in protein misfolding diseases (e.g., Prion diseases).~~

~~The use of any one of claims 120-122, wherein the disease or disorder is a neurodegenerative disease; optionally, the disease or disorder is Parkinson's disease or Alzheimer's disease.~~

~~The use of any one of claims 120-122, wherein the disease or disorder is a dermatological disease; optionally, the disease or disorder is acne.~~

~~The use of any one of claims 120-122, wherein the disease or disorder is cancer; optionally, the cancer is metastasising cancer, brain cancer, gastrointestinal cancer, skin cancer, non-small-cell lung carcinoma, head and neck squamous cell carcinoma or colorectal~~